Chapter 1

Java Introduction

* Java is a high-level language & object-oriented.
* Language → way of instruction (communication layer)
* Types : → (i) High-level
  (ii) Middle-level
  (iii) Low-level
* Regular 6-month updation regulations.
* Compiler checks syntax errors / compiler time error.
* Interpreter checks runtime error.
* JVM → compilation } → JDK
  ↓ → runtime } → JRE → JVM
  (Interpreter part mostly).

* Bytecode → .class file
  Java → machine language
  JIT → used for fast execution
  JRE → integreated JVM / libraries.
  JDK → JRE + JVM

* JRE → consists of necessary libraries for running.
* JVM → executes the program.

* IDK → Integrated Development kit.
  → Programmer usable kit.

* Magic Bytecode : special unique code. (CAFEBABE)
  - used for the understanding of java file recognition (class file). without this recognition of java file is not possible.

Chapter - 2
Structure / Data types / (Programming Fundamentals

* Case - sensitive language
  MyClass.java ≠ myclass

* Public class → must have atleast one class & function of "main".
* Every statement must end with (;).

* Data types :
  → int, double (decimal), char, boolean (0/1), string, float (0.1), etc.

(i) Primitive	Non-primitive
- 8 types, fixed size.	- non-fixed variable
- Eg : Byte, int, char	- eg : String, Class, array.
- 1 Byte → 8 bit → -8 to 128	- Sequence, collection

* definition → closed by ;
* declaration → using assignment operator (=).

* Variable Types :
  (i) Local : declared inside the function.
  (ii) Instance : object ~~can be~~ are accessed through this (copies of variables)
  (iii) Static : without object.

* String [] args → main function of arguments.

* Type casting : converts one type of data to another
  Types :
  (i) Widening (Automatic / implicit)
  ↳ small → large (no data loss)
  (ii) Narrowing (Explicit / Manual)
  ↳ Large → Small (data is truncated)

(smallest)
↑
byte → short → int → long
↓
double ← float
↘ (biggest)

* Operators : arithmetic, logical, relational, urinary, etc
* Scanner : to take user input.
  → uses util.scanner
  ↳ syntax : import java.util.scanner.
  → util is the package here.